MSBA 6131 (Fall 2025) Live Case

In Collaboration with CSI

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1. Problem Definition

This will be discussed in detail during the live case kickoff event by CSI. Broadly speaking, CSI keeps records of historical auctions of construction projects and the corresponding bids in those auctions. The objective is to organize the bidding data in a meaningful manner and explore possible insights that can eventually inform CSI's bidding strategies.

2. Data Description

The live case uses two data files, provided by CSI, that cover auction and bidding information in 2024 and 2025. Data file will be shared on Canvas. See below for explanation of each field (fields that are self-explanatory are omitted here):

- **Mix TN**: The total tons of asphalt mix used on the project. "Mix" refers to the hot mix asphalt (HMA) that forms the finished roadway surface.
- **Grade CY**: The total cubic yards of material to be moved during grading operations. This represents the earthwork required to shape the roadbed, typically performed using dozers, motor graders, or scrapers.
- **Base TN**: The total tons of base material placed on the project. The base layer is installed beneath the asphalt surface and provides structural support for the pavement.
- **SHLDR TN**: The total tons of gravel placed on the roadway shoulders after the hot mix asphalt has been laid. Shoulders provide lateral support and drainage along the edge of the pavement.
- **Mill SY**: The total square yards of existing pavement to be milled or removed. Milling involves grinding off the old road surface in preparation for new asphalt placement.
- **Crush TN**: The total tons of aggregate material to be produced for the project. Material is made by processing gravel from pits through a crusher to create the necessary sand and rock sizes.
- **Stock TN**: The total tons of material to be stockpiled for later use on the project. Stockpiled materials may include aggregate, base, or recycled asphalt.

- **Box**: Indicates whether the project includes a box culvert. (Yes 'X'/No nothing noted)
- **Prevailing Wages**: indicates whether prevailing wage requirements apply to the project. Prevailing wages are the minimum hourly wage rates that must be paid to workers on public construction projects funded by State or Federal sources. These rates are set by government agencies (such as the U.S. Department of Labor or state departments of transportation) and are based on surveys of local wages and fringe benefits for each trade classification (e.g., laborers, operators, truck drivers). Contractors and subcontractors are legally required to pay at least the prevailing wage rates in effect at the time of bid.
- **Diesel prices**: Average monthly diesel prices.
- **Bidder_i_Name** and **Bidder_i_Total**: the name and total bid amount of a particular bidder (where *i* is the index). Please note that
    - In most cases, a bidder with smaller index has a lower bid amount (meaning that the bidder is willing to take the contract with a lower payment), but not always (due to data quality issues). You are recommended to re-sort them as part of your data processing step.
    - Most bidders are construction companies (you can tell from their names) with one exception called "engineer estimates", which not an actual bidder but instead represents the official estimates of the project costs. It is often used as a "reasonable cost projection" to benchmark submitted bids – if you bid wildly higher or lower than the engineer estimates, your bid may not be accepted.

3. Some Suggestions

Below are a few suggestions for you to consider as you construct your own solutions.

Clean the Data before going any further.

Despite hundreds of hours of manual labor and the help of a dozen volunteers (myself included) to manually process and clean this data, it is still not clean. You can largely trust the numerical bid amounts, but there are many duplicates in bidder names (that is, two bidders that are actually the same company despite having different names). For example, "JENSEN BROS CONST" and "JENSEN BROTHERS CONSTRUCTION INC" are actually the same construction company. This is a vivid example of the "data consistency" problem we have discussed in class. Please

apply your own common sense judgement to further clean the datasets. When in doubt, reach out to me or the client.

Think deeply about what "perspective" you want to focus on.

Please keep in mind that this is an "exploratory" project, where the objective is to identify potentially interesting insights / patterns from the data (without knowing what might be "interesting" beforehand). Due to its inherent ambiguity, it is very important to have a clear focus, rather than trying to explore everything (which isn't feasible). Ultimately, the client wants to increase their success rate in project bidding – that is, they want to win more contracts at a reasonable price (i.e., not having to aggressively overbid). From this end goal, try to come up with your own perspectives on how to inform that ultimate goal.

Modeling techniques and programming tools.

In general, you are not restricted to using only the techniques / methods we covered in class. Furthermore, you have flexibility in terms of programming language (R or Python). Your choice of programming language has no impact on your live case grades.

Deliverable (important!!!)

The final deliverable is a dashboard and a set of presentation slides that communicate your findings and allow the client to explore. Notice this doesn't mean that you have to do all your work within tools like Tableau. For example, it's perfectly fine (and necessary) to carry out data pre-processing / analyses in R, and then build a dashboard in Tableau to visualize them. Please *DO NOT* publish your Tableau Dashboard on Tableau Public – we will coordinate with the client about knowledge transfer at the end.

4. Live Case Grading (35% of Total Course Grade)

Your performance on the live case is determined based on an internal presentation that will be graded by Lee and me (accounting for 28 pts). Additional information about the internal presentation, including the submission deadline, submission format, and other guidance materials, will be announced on Canvas. Around 6-7 top teams will advance to present in front of the client. The winning team(s) will receive a *non-consequential* bonus to their live case grades.

Before the midpoint Q&A session with client, each team should also submit a list of questions they would like to ask. This accounts for 2 pts.

Finally, teamwork is of critical importance in this live case, and 5 pts of the live case grades will be allocated based on team performance. I will assess team performance based on following dimensions (adapted from Svjetlana's "Building & Managing Teams" workshop):

- Working Together: are team members able to work together effectively?
- Role Clarity: are team roles clearly defined (note the difference with technical roles)?
- Respect and Diversity Appreciation: do team members respect each other and appreciate the diverse background / skills of each member?
- Active Listening: are team members able to listen actively to each other during meetings?
- Addressing Weak Links: is the team able to effectively identify and address any weak links in collaboration?
- Cohesion: generally speaking, is the team operating in a cohesive way?